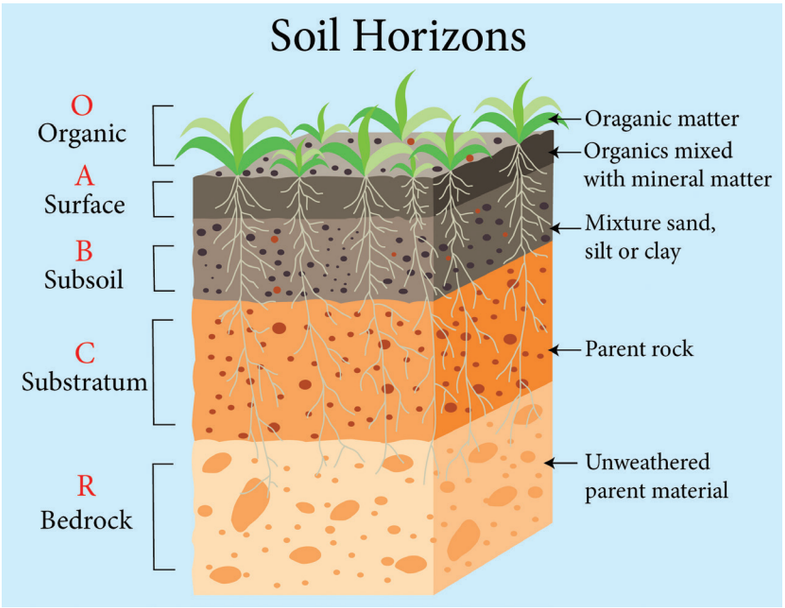


Figure 10.1: Soil Horizons

10.1.2 Soil Horizons

Each type of soil is characterized by the presence of different horizons which can be seen in a soil profile (Figure 10.1). The formation of soil horizons depends on climate, living organisms, parent rock material, topography and time; all of which control the weathering of rocks.

10.1.3 Physical and Chemical Properties of Soil

Physical properties of a soil type depends on the size of particles, soil texture, soil temperature and soil pH.

Chemical properties of soil includes three main components which provides nutrients for plant growth. The three components are the organic matter, the derivatives of parent rock materials and the clay fraction.

The fertility of soil depends not only on its chemical composition, but also on the qualitative and quantitative nature of microorganisms inhabiting it. The branch of science dealing with study of soil microorganisms and their activities in soil is known as '**Soil Microbiology**'.

10.2 Pioneers of Soil Microbiology

Scientists studied the microorganisms from water, air and soil. They recognized the role of microorganisms in natural processes. They realized the importance of soil microorganisms in growth and development of plants. Soil Microbiology emerged as a distinct branch of soil science during first half of the 19th century.

Sergei N. Winogradsky discovered the autotrophic mode of life among bacteria and established the microbiological transformation of Nitrogen and Sulphur. He isolated nitrifying bacteria for the first time and demonstrated the role of these bacteria in nitrification (1890). Further he demonstrated that free-living *Clostridium pasteurianum* could fix atmospheric Nitrogen (1893). He developed the Winogradsky column (Figure 10.2), a self contained ecosystem for studying the

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